

Activity 14

STAT 184

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Table of contents

0.1	Introduction	1
0.2	Armed Forces Data Wrangling Redux (Activities #08 and #10)	1
0.3	Popularity of Baby Names(Activity #13)	2
0.4	Plotting a Mathematical Function(Activity #4)	3
0.5	Reflection	3
0.6	Code Appendix	4
1	Activity #8:	4
2	Activity #10:	5
3	Activity #13:	7
4	Activity #4:	8

0.1 Introduction

This document contains the R code for activities 4, 8, 10, and 13 regarding armed forced data wrangling, popularity of baby names, and plotting mathematical functions.

0.2 Armed Forces Data Wrangling Redux (Activities #08 and #10)

Table 1: Preview of Individual Soldier Data

Pay.Grade	Branch	Sex	Rank
-----------	--------	-----	------

E1	Army	Male	Private
E1	Army	Male	Private
E1	Army	Male	Private
E1	Army	Male	Private
E1	Army	Male	Private
E1	Army	Male	Private

0.3 Popularity of Baby Names(Activity #13)

I picked the names Emma, Noah, Olivia and Ethan since they were some of the most popular names I could think of. The visualization displays and compares the popularity of female and male names from 1880 to 2020. We can see that all 4 names became much more popular in the last 50 years.

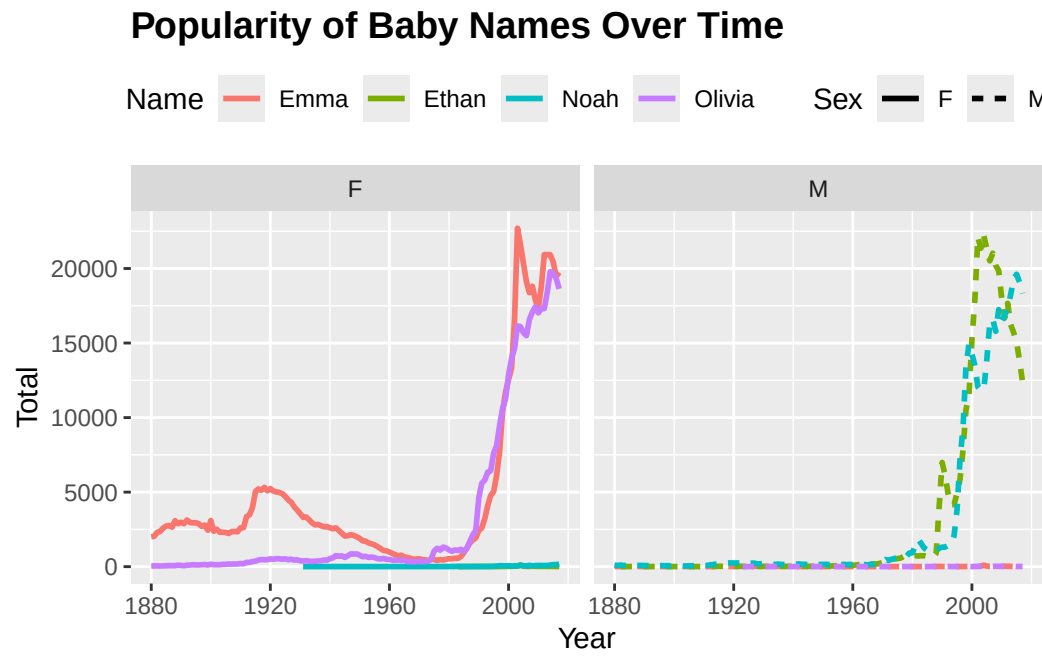
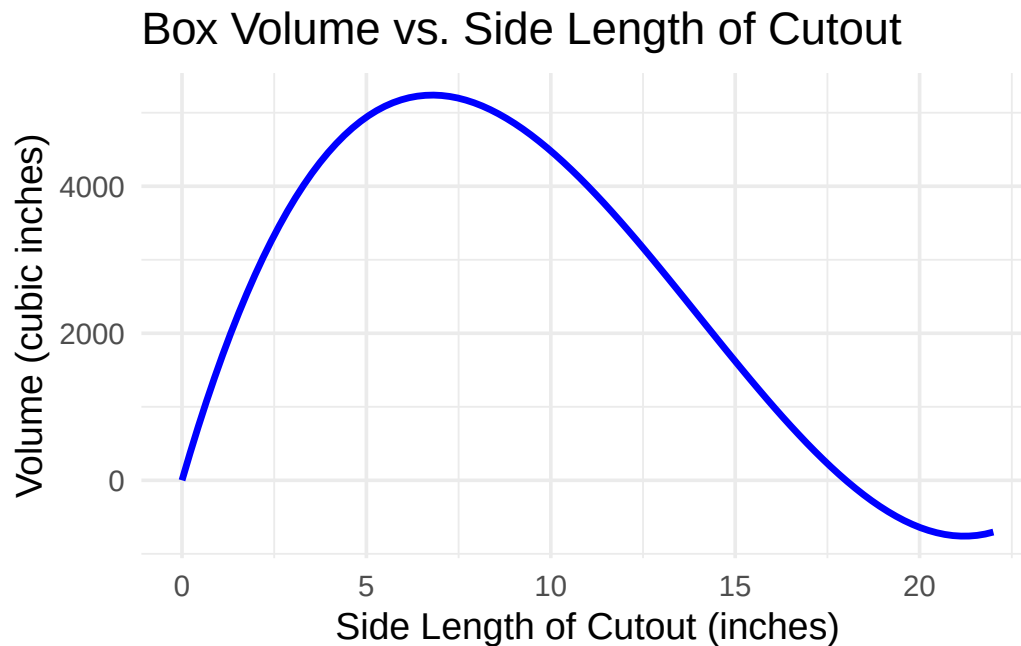


Table 2: Table 1. Total Number of Army Warrant Officers by Sex

Male	Female
514	1564
Total	1564

0.4 Plotting a Mathematical Function(Activity #4)

This plot shows how the total volume of an open-top box changes as the side length of the cutout increases from 0 to 22 inches. The volume increases sharply until it hits its peak at about 7 inches, and then it starts decreasing. The plot shows how optimizing the cut size leads to the largest possible volume for a 36 by 48 inch sheet



0.5 Reflection

So far in this course, I have learned a lot about the R programming language. It started off with simple plots based on equations like the one from activity #4. Then, I learned how to use packages from R and use datasets found in those packages to create more complicated data visualizations using more complex functions of R. I also learned how to create visualizations with data found on the internet or from files downloaded from my computer. Finally, I have now learnt how to create Quarto markdown files from doing this project. Overall, this class has been a great learning experience with the R programming language.

0.6 Code Appendix

1 Activity #8:

```
library(tidyverse)
library(rvest)
library(googlesheets4)

rank_page <- "https://neilhatfield.github.io/Stat184_PayGradeRanks.html"

ranks_raw <- rank_page %>%
  read_html() %>%
  html_elements("table") %>%
  html_table() %>%
  pluck(1)
#clean rank data
ranks_raw[1, 1] <- "Type"
colnames(ranks_raw) <- ranks_raw[1, ]
ranks_trimmed <- ranks_raw[-c(1, 26), ]

ranks_tidy <- ranks_trimmed %>%
  select(-Type) %>%
  pivot_longer(
    cols = -`Pay Grade`,
    names_to = "Branch",
    values_to = "Rank"
  ) %>%
  mutate(Rank = na_if(Rank, "--"))

gs4_deauth()
#read in the armed forces data from Google docs
sheet_url <- "https://docs.google.com/spreadsheets/d/19xQnI1cBh6Jkw7eP8YQuuicM1VDF7Gr-nXCb5q"

headers_preview <- read_sheet(sheet_url, col_names = FALSE, n_max = 3)
#read the data while skipping headers
forces_raw <- read_sheet(
  sheet_url,
  col_names = FALSE,
  skip = 3,
  n_max = 28,
  col_types = "c"
```

```

)
#Cleaning the forces data
branches <- c("Army", "Navy", "Marine Corps", "Air Force", "Space Force", "Total")
expanded_branches <- rep(branches, each = 3)
col_names <- paste0(c("", expanded_branches), ".", headers_preview[3, ])

names(forces_raw) <- col_names

forces_tidy <- forces_raw %>%
  rename(Pay.Grade = `.Pay Grade`) %>%
  select(-contains("Total")) %>%
  filter(!Pay.Grade %in% c("Total Enlisted", "Total Warrant Officers", "Total Officers", "Total")) %>%
  pivot_longer(
    cols = -Pay.Grade,
    names_to = "Branch_Sex",
    values_to = "Frequency"
  ) %>%
  separate(Branch_Sex, into = c("Branch", "Sex"), sep = "\\.") %>%
  mutate(
    Frequency = na_if(Frequency, "N/A*"),
    Frequency = parse_number(Frequency)
  )
#merge rank data and forces data
forces_with_ranks <- forces_tidy %>%
  left_join(ranks_tidy, by = c("Pay.Grade" = "Pay Grade", "Branch" = "Branch"))
#create data frame with individual data
soldiers_individual <- forces_with_ranks %>%
  drop_na(Frequency) %>%
  uncount(weights = Frequency)
head(soldiers_individual) %>%
  knitr::kable(caption = "Preview of Individual Soldier Data") %>%
  kableExtra::kable_styling(full_width = FALSE)

```

2 Activity #10:

```

library(janitor)
library(knitr)
library(kableExtra)
library(dplyr)

```

```

library(tidyr)

# Reads in data, adjust file path if needed
military_data <- read.csv("C:\\Users\\alfre\\Downloads\\US Armed Forces (6_2024) - Sheet1.csv",
                        skip = 1, header = TRUE)

# Rename columns
colnames(military_data) <- c(
  "Pay_Grade",
  "Army_Male", "Army_Female", "Army_Total",
  "Navy_Male", "Navy_Female", "Navy_Total",
  "Marine_Male", "Marine_Female", "Marine_Total",
  "AirForce_Male", "AirForce_Female", "AirForce_Total",
  "SpaceForce_Male", "SpaceForce_Female", "SpaceForce_Total",
  "Total_Male", "Total_Female", "Total"
)

# Clean data
military_data <- military_data %>%
  filter(!grepl("Source|Pay Grade|Total", Pay_Grade))

# Convert columns to be numeric
military_data[, -1] <- lapply(military_data[, -1], as.numeric)

# Choosing which branch
chosen_branch <- "Army"
branch_data <- military_data %>%
  select(Pay_Grade,
         Male = paste0(chosen_branch, "_Male"),
         Female = paste0(chosen_branch, "_Female"))

branch_data <- branch_data %>%
  mutate(Rank_Category = case_when(
    grepl("^E", Pay_Grade) ~ "Enlisted",
    grepl("^W", Pay_Grade) ~ "Warrant Officer",
    grepl("^O", Pay_Grade) ~ "Officer",
    TRUE ~ "Other"
  ))

# Filter only Warrant Officers
branch_data <- branch_data %>%
  filter(Rank_Category == "Warrant Officer")

```

```

# Sum up counts for each gender
freq_table <- branch_data %>%
  summarise(
    Male = sum(Male, na.rm = TRUE),
    Female = sum(Female, na.rm = TRUE)
  ) %>%
  adorn_totals("row")

# Frequency table
freq_table %>%
  kbl(caption = paste("Table 1. Total Number of", chosen_branch, "Warrant Officers by Sex"))
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover"))

```

3 Activity #13:

```

# ALT TEXT: Line graph of baby name popularity over time for Emma, Olivia (female) and Ethan

library(babynames)
library(dplyr)

names <- c("Emma", "Noah", "Olivia", "Ethan")
#Filters out names that aren't the selected ones
babynames_filtered <- babynames %>%
  filter(name %in% names) %>%
  group_by(year, name, sex) %>%
  summarise(total = sum(n), .groups = 'drop')

ggplot(babynames_filtered, aes(x = year, y = total, color = name, linetype = sex)) +
  geom_line(size = 1) +
  labs(
    title = "Popularity of Baby Names Over Time",
    x = "Year",
    y = "Total",
    color = "Name",
    linetype = "Sex"
  ) +
  #Creates two separate graphs
  facet_wrap(~sex) +
  theme(

```

```

    plot.title=element_text(face="bold", size=14),
    legend.position="top"
)

```

4 Activity #4:

#ALT TEXT: Graph showing Box Volume vs. Side Length of Cutout with a blue curve that peaks near 11 inches.

```

library(ggplot2)

volume <- function(x, length = 36, width = 48) {
  (length - 2*x) * (width - 2*x) * x
}

ggplot(data.frame(x = c(0, 22)), aes(x = x)) +
  stat_function(fun = volume, color = "blue", linewidth = 1.2) +
  labs(
    title = "Box Volume vs. Cut Size",
    x = "Side Length of Cutout (inches)",
    y = "Volume (cubic inches)"
  ) +
  theme_minimal(base_size = 14)

```